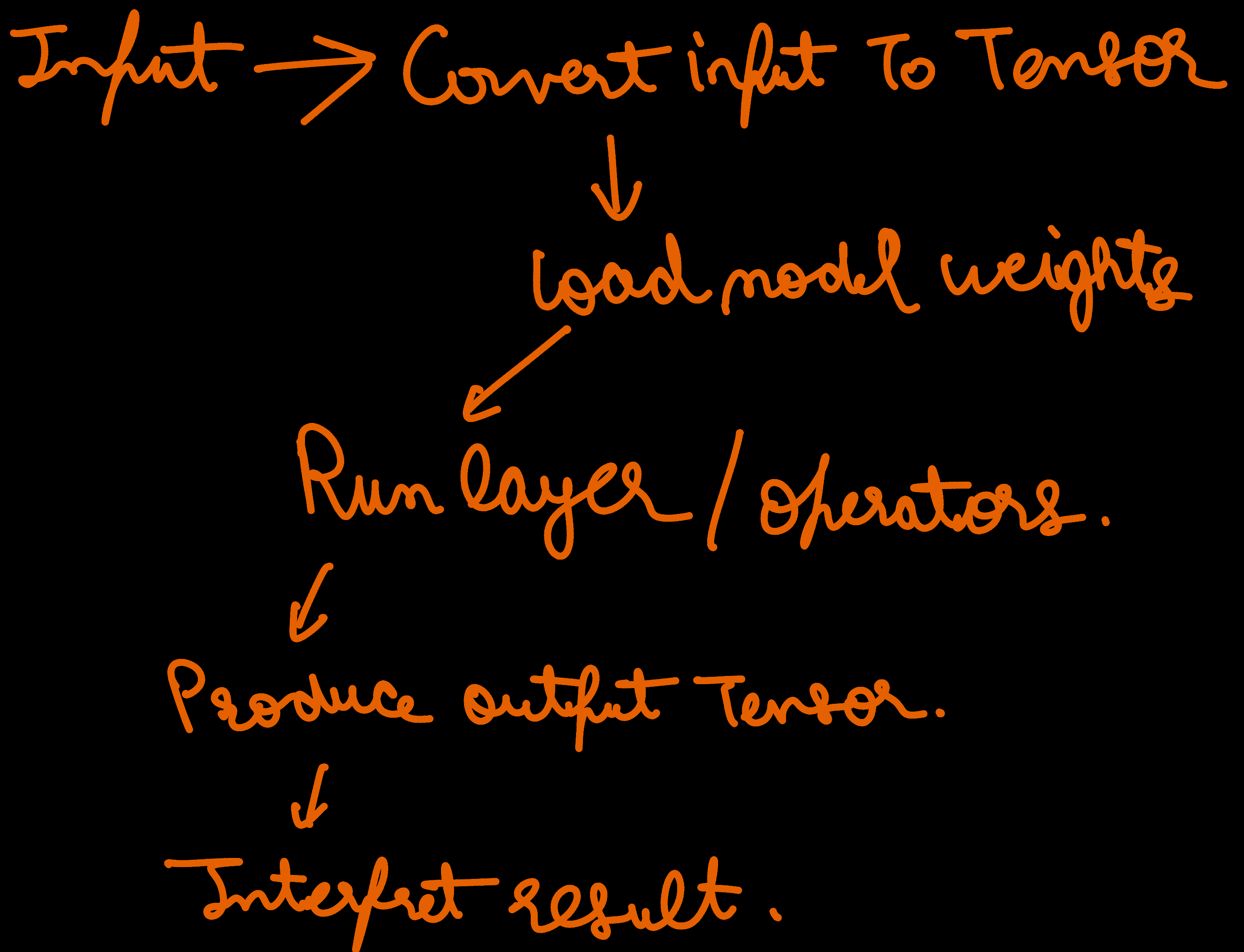


A Small CPU runtime that
executes a pretrained model
by moving Tensors Through numerical
Operation.



Modules

→ Tensor → numerical data
memory layout.
dimension.

Used by everything.

Operator module

→ To Compute.

and give back updated

Tensor.

③ Engine runs
everything

Runtime itself usually does NOT:

perform math directly

It CALLS operators.

That separation matters.

④ Load the
weight.

⑤ Benchmark.

⑥ Input
module
also

(Image)
Tensor.

Note Tensor is not just an Input Container, its every Single Contains weights, input, output everything is Tensor. So Tensor needs to be general.

weights are tensors
 activations are tensors
 outputs are tensors
 inputs are tensors

for a Gemm it only know
 how to deal with Tensors.

i Jk did 45% better than i k J ?

CPU
 load
 Sum

load
 modify
 update

in loop

Sum
update

which was
bad

- Even if the value stays in cache, that's still more traffic than:

single local
accumulator